MIL-F-38008D(USAF)
14 August 1969
Superseding
MIL-F-38008C(USAF)
6 December 1967

MILITARY SPECIFICATION

FILTER ELEMENT, FLUID PRESSURE MXU-234/E

1. SCOPE

1.1 This specification covers one type of fluid pressure filter element, designated MXU-234/E.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (For Carton Sealing)

Military

MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-G-5572	Gasoline, Aviation, Grades 80/87, 100/130, 115/145
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-F-8901	Filter-Separators, Aviation and Motor Fuel, Ground and Shipboard Use, General Requirements and Test Procedures for
MIL-P-19834	Plates, Identification, Metal Foil, Adhesive Backed

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC 4330

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MIL-STD-130 Identification Marking of U.S. Military Property
MIL-STD-831 Test Reports, Preparation of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

D256 Standard Methods of Test for Impact Resistance of Plastics and Electrical Insulating Materials

D638 Method of Test for Tensile Properties of Plastics

D790 Standard Method of Test for Flexural Properties of Plastics

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 General specification. The requirements of MIL-F-8901 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.3 Materials

3.3.1 Metallic parts. All metallic parts of the filter element shall be fabricated from an aluminum alloy suitable for the application.

3.3.2 Fungusproof materials. Materials that are not nutrients for fungi shall be used to the greatest extent practicable. Where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent.

3.3.3 Petroleum products. The materials used in the construction of the filter element shall neither affect nor be affected by petroleum products.

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3.3.4 Plastic parts. If used in the construction of the elements, molded plastic parts shall have the following physical properties:

a. Tensile strength, minimum - 4,500 psi

b. Flexural strength, minimum - 7,500 psi

c. IZOD Impact, minimum - 4.5 pounds per inch of notch.

3.4 Design and construction. The filter element shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the stresses, jars, vibrations, and other conditions incident to shipping, storage, installation, and flight line service.

3.4.1 Functional design. The filter element shall be designed for installation in Warner Lewis Company M-396 and M-397 filter-separator conversion kits. The filter element shall be designed as shown on figure 1.

3.4.1.1 Compatible fuels. The filter element shall be so designed that it can be used with fuels conforming to MIL-G-5572 and MIL-T-5624.

3.5 Performance. When 10 elements are installed in a Faudi Feinbau Company model FWE-12-10 filter-separator as shown on figures 2 and 3, the filter element shall be capable of meeting the performance requirements of MIL-F-8901 at a fuel flow rate of 600 gpm, except that the influent fuel shall contain water only up to 3 percent by volume.

3.6 Identification of product. The following information shall be marked on the filter element in accordance with MIL-STD-130. If the filter element is marked on the sealing surface, the markings shall not affect the sealing of the element.

Manufacturer's part No._____
Manufacturer's name or trademark_____
Date of manufacture_____

3.6.1 Manufacturer's part number. The manufacturer's part number and drawing number shall be the same.

3.6.2 Nameplate. A nameplate fabricated from materials as specified in MIL-P-19834, type 1, shall be supplied one per each five elements shipped. The nameplate shall be attachable by Government personnel to the pressure vessel to facilitate maintenance of the fueling system. Attachment instructions shall be furnished, attached to the plate. The nameplate shall conform to figure 4. When specified (see 6.2), additional nameplates shall be provided.

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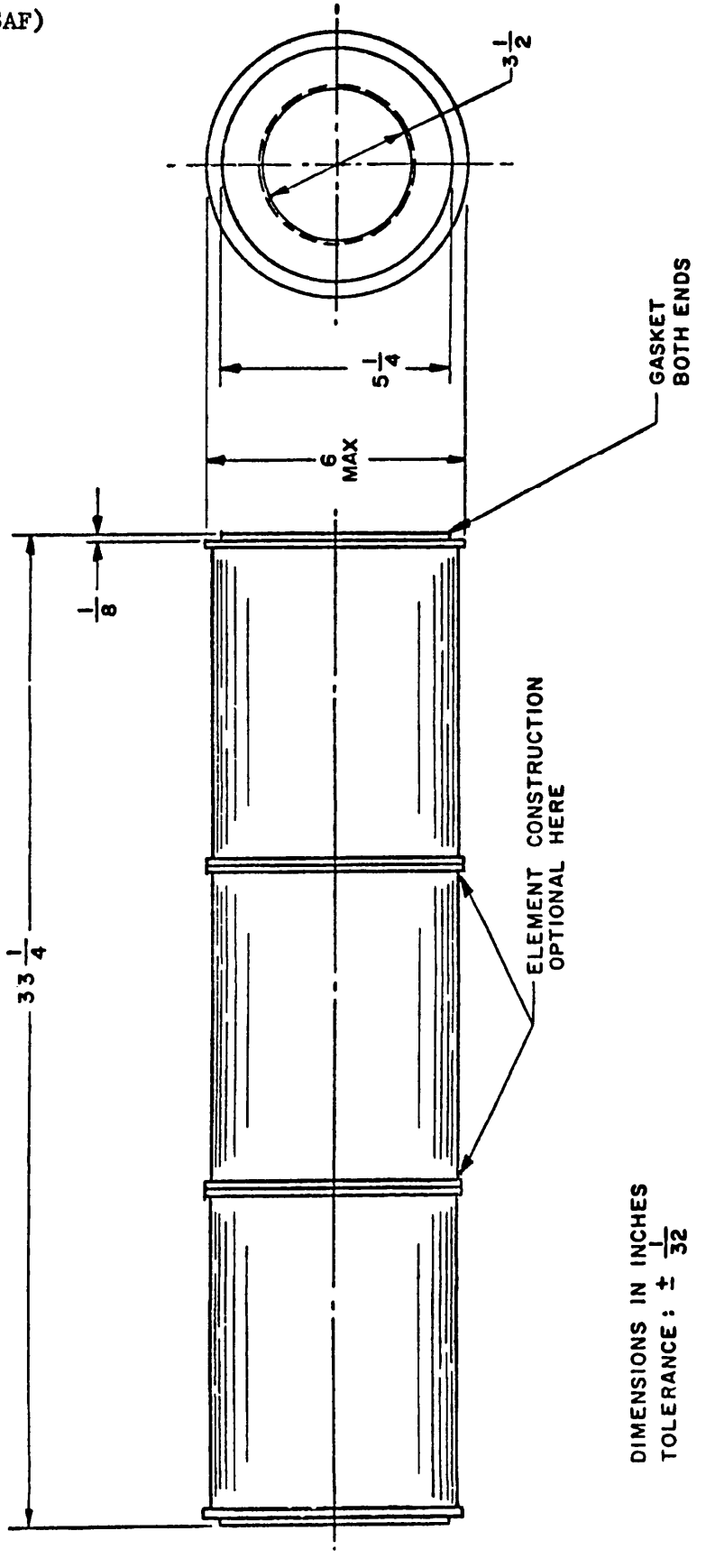


FIGURE 1. Filter Element

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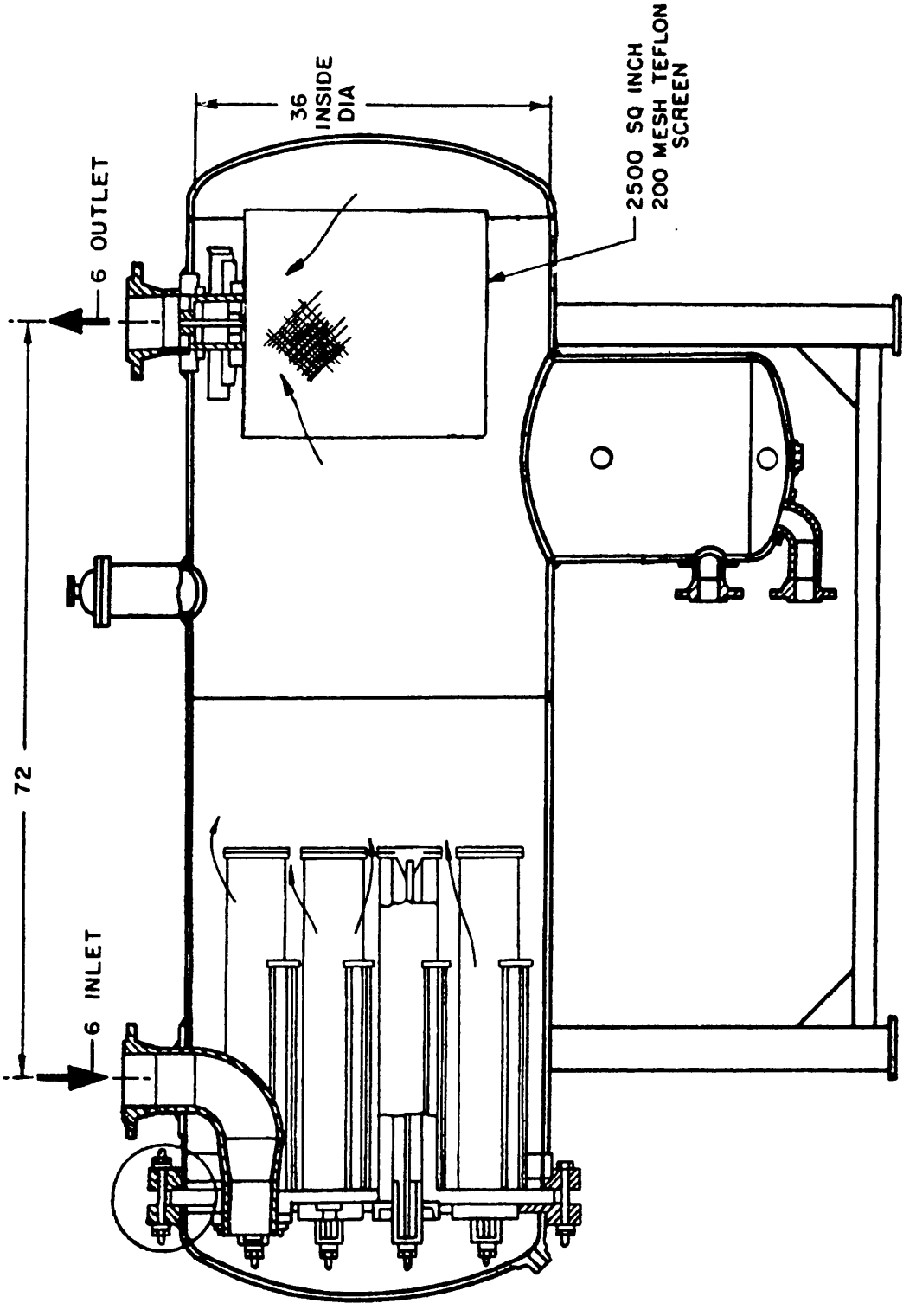


FIGURE 2. Side View

DIMENSIONS IN INCHES

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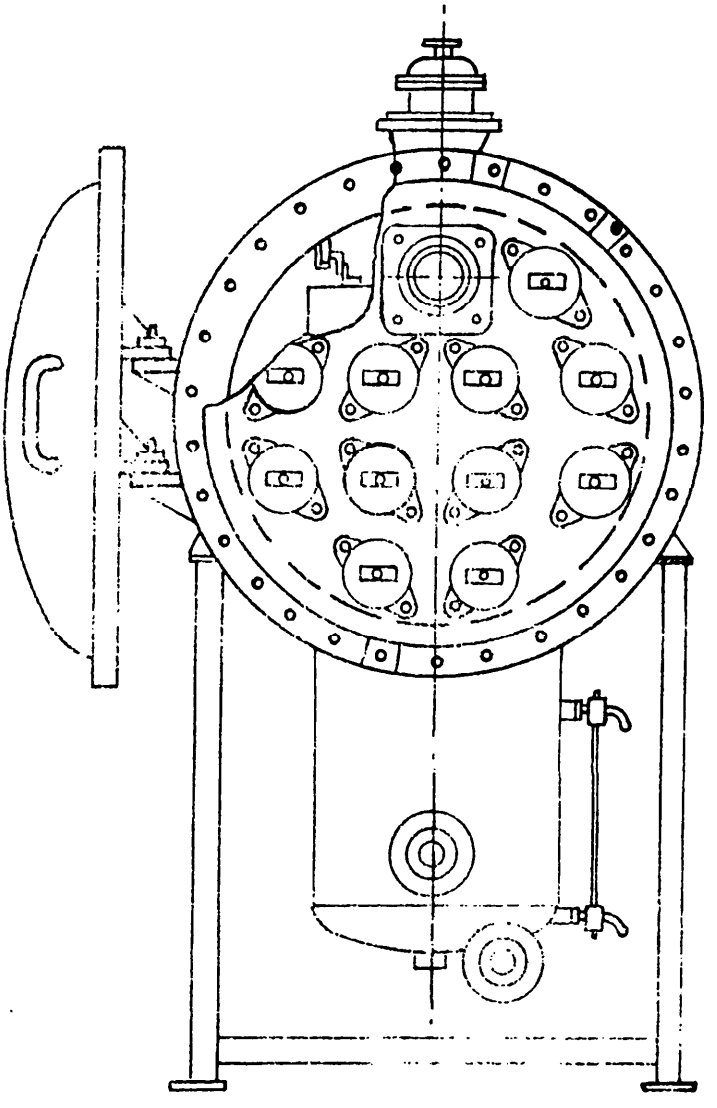
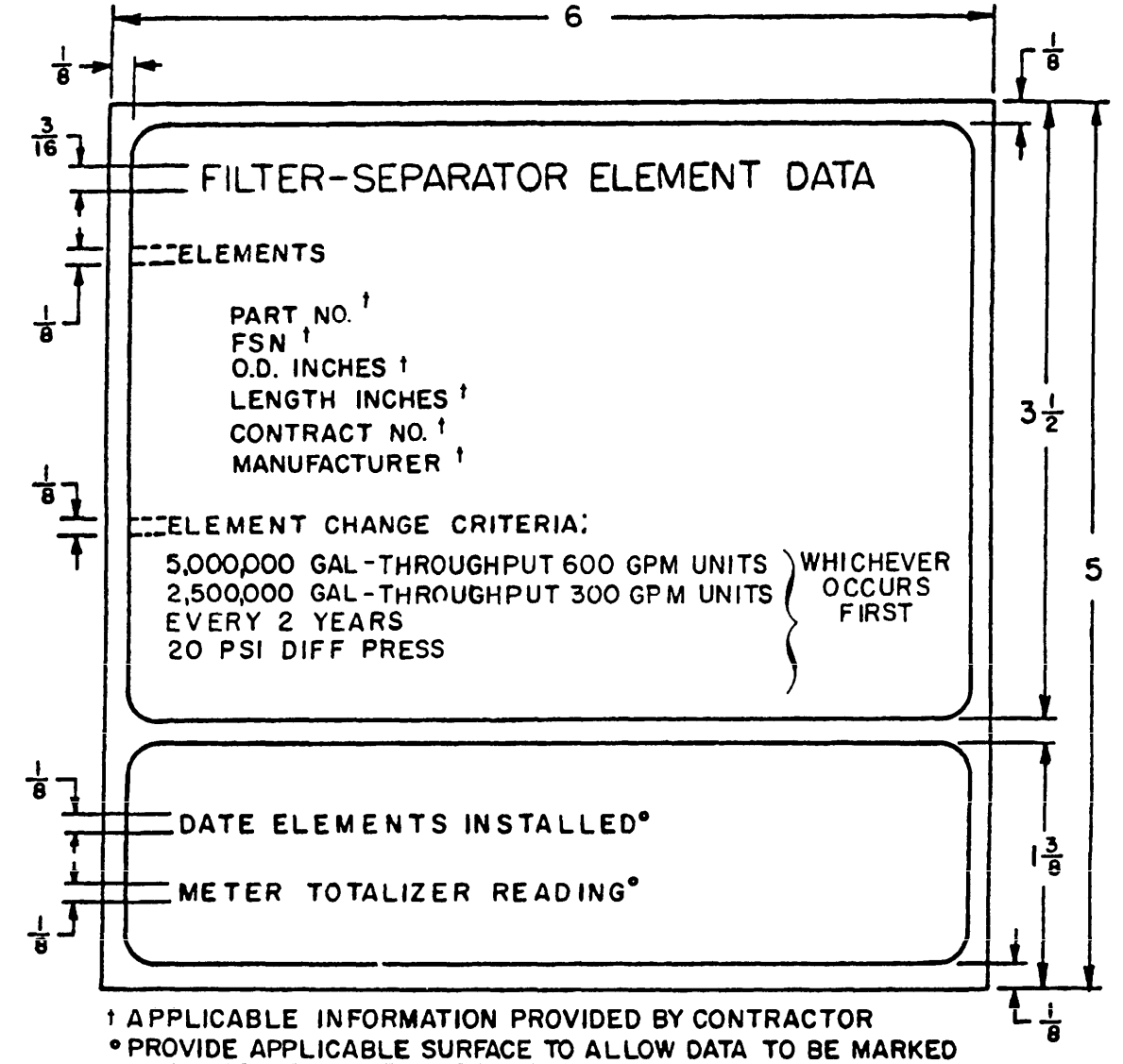


FIGURE 3. View of Opened Cover

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† APPLICABLE INFORMATION PROVIDED BY CONTRACTOR
° PROVIDE APPLICABLE SURFACE TO ALLOW DATA TO BE MARKED IN BY GOVERNMENT PERSONNEL

NOTES:

1. ALL LETTERING SHALL BE 3/32-INCH HIGH, UNLESS OTHERWISE NOTED
2. BACKGROUND SHALL BE BLUE
3. DIMENSIONS IN INCHES, TOLERANCE ± 1/32

FIGURE 4. Nameplate

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3.7 Workmanship. The filter element, including all parts, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection and testing. The inspection and testing of the filter element shall be conducted in a full-scale filter-separator unit in accordance with MIL-F-8901 and as specified herein.

4.2 Test condition. During all applicable full-scale tests, 10 filter elements shall be mounted in a Faudi Feinbau Company filter-separator housing as shown on figures 2 and 3. During all functional testing, 2 of the 12 element mounting holes shall be plugged to prevent fuel bypassing.

4.3 Preproduction testing (see 6.2)

4.3.1 Test samples. Unless otherwise specified (see 6.2), the preproduction test samples shall consist of 44 elements representative of a production lot (see 6.3.1). The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity. Samples shall be prepared as follows:

4.3.1.1 Data to accompany test samples. The preproduction test samples shall be accompanied by the following data:

a. Brief operating and installation data to enable test personnel to correctly install and operate the equipment

b. Engineering data in the form of assembly drawings (two sets).

4.3.1.2 Test sample and data for the procuring activity. When tests are conducted at a site other than the laboratory of the procuring activity, the following shall be furnished that activity:

a. Three copies of a test report in accordance with MIL-STD-831. In addition to results, the report shall contain complete records of the tests, including data sheets, performance curves, chronological test records, photographs, sample calculations, test procedures, and a description of the test apparatus.

b. A sample element from the filter-separator which was subjected to the preproduction tests.

c. Color photograph of the sample element.

d. A new filter element.

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4.3.2 Preproduction tests. The preproduction tests shall consist of the applicable tests specified in MIL-F-8901 except or modified as follows:

a. The Hydrostatic pressure test shall not be conducted.

b. The Fuel shutoff control test shall not be conducted.

c. The Water removal tests shall be performed at a total input rate of 600 gpm. The water injection for the second hour shall be 3 percent by volume in lieu of 5 percent. The third hour is not required.

d. The Inhibited fuel test may be accomplished using a recirculating-type system.

e. The electric chart recorder shall be used with the Totamitor.

f. Sections of the plastic material used in the construction of the element end caps or specially prepared test specimens shall be tested as indicated below to verify plastic material compliance. The resultant test data shall be included in the test report.

(1) Tensile strength - ASTM D638, type 1

(2) Flexural strength - ASTM D790

(3) IZOD impact strength (notched) - ASTM D256, method A.

4.4 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests.....See 4.4.1

b. Sampling plan and test..See 4.4.2

4.4.1 Individual tests. Each filter element shall be examined as required by the Examination of product paragraph of MIL-F-8901, and to determine that it conforms to the filter elements used in the preproduction tests.

4.4.2 Sampling plan and test. Samples of the filter element shall be selected at random from each lot in accordance with MIL-STD-105 and subjected to the Filter/coalescer element test specified in MIL-F-8901. An AQL of 1.5 percent defective, level I, shall be established. This test shall be considered destructive.

4.4.3 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

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5. PREPARATION FOR DELIVERY

5.1 Packaging (see 6.2)

5.1.1 Level A. One filter element shall be packed in accordance with method III of MIL-P-116. Each filter shall be wrapped in grade A barrier material conforming to MIL-B-121, secured with tape conforming to PPP-T-76, and placed in a container conforming to PPP-B-636.

5.1.1.1 Nameplate. Each nameplate shall be packaged in accordance with method III of MIL-P-116 by wrapping in grade A barrier material conforming to MIL-B-121 and securing with tape.

5.1.1.2 Intermediate packaging. Ten unit-packaged filters, and two unit-packaged nameplates, shall be intermediate packaged in a container conforming to PPP-B-636.

5.1.2 Level C. Preservation and packaging shall be the minimum necessary to afford protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity. One nameplate (see 3.6.2) shall be packaged in the same carton with each five elements or fraction thereof. The other nameplates shall be shipped per instructions of the contracting officer (see 6.2).

5.2 Packing (see 6.2)

5.2.1 Level A. Filter elements shall be packed in overseas-type containers conforming to PPP-B-621 or PPP-B-601. Containers shall be of minimum weight and cube, uniform in size and shape, and shall contain like quantities, when practical. Closure of containers shall be in accordance with the applicable specifications and appendix thereto.

5.2.2 Level B. Filter elements shall be packed as specified in 5.2.1, except containers shall be of the domestic-type and closure shall be in accordance with the requirements for the domestic containers.

5.2.3 Level C. Packages which require overpacking for acceptance by the carrier shall be packed in shipping containers in a manner that will insure safe delivery at the lowest transportation and packaging charges. Containers shall meet Consolidated freight rules and regulations or other rules and regulations applicable to the mode of transportation.

5.3 Shipment marking. In addition to any special marking required by the contract or order (see 6.2), all shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

FILTER ELEMENT, FLUID PRESSURE MXU-234/E.

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6. NOTES

6.1 Intended use. The MXU-234/E filter element is intended for use as a replacement item in ground aviation liquid fuel filtering and servicing systems. This filter element is intended to replace Warner-Lewis Company part No. C-E-3 elements in Warner-Lewis Company model M-396 and M-397 filter-separator conversion kits.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Location and conditions for preproduction testing (see 4.3)

c. When the test samples are to be other than as specified in 4.3.1

d. Applicable levels of preservation, packaging, and packing (see 5.1 and 5.2)

e. The quantity of additional nameplates required (see 3.6.2) and shipping instructions for them (see 5.2.1 and 5.2.3)

f. Special shipment marking (see 5.3).

6.3 Definition. For the purpose of this specification, the following defifition will apply:

6.3.1 Lot. All filter elements of the same part number manufactured in a production run and submitted for acceptance at the same time shall be considered a lot for acceptance purposes.

6.4 Contract administration

6.4.1 The contracting organization should require that a prospective contractor shall have, or can contract for, a suitable test system prior to the award of any contracts for filtration equipment. A test system is the piping, test requirement apparatus, arrangement of test components, laboratory equipment, laboratory supplies, and techniques used during the evaluation of fuel filtration equipment.

6.4.2 The contracting organization should provide for Government inspection during the preproduction buildup and test in addition to providing Government inspection after production is released.

6.4.3 The organization should, prior to issuance of a procurement bid for proposal (IFB/RFP), contact the engineering activity having technical responsibility for the specification to confirm that the latest edition of the specification is being cited.

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6.5 Abbreviations. In order to save space and to avoid the spelling out of repetitious words and phrases, the following abbreviations have been used in the text of this specification:

AQL ..... acceptable quality level

ASTM .... American Society for Testing and Materials

dia ..... diameter

gal ..... gallon

gpm ..... gallons per minute

OD ..... outside diameter

psi ..... pounds per square inch

6.6 Identification of changes. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 11

Reviewer:
Air Force 84

Preparing activity:
Air Force - 11

Project No. 4330-F081

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

AFLC–WPAFB–OCT 67 2M

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